Budgeted OpenMP Scheduling Selection: Time, Compute, and Execution-Quality Trade-offs

Shaan Chopra, Mrinal Dhami, and Aniket Verma
School of Computer Science and Engineering
Manipal Institute of Technology
Group 3: 240905140, 240905026, 240905156

***Abstract*—Finding the fastest loop scheduler is useful only when the improvement repays the cost of finding it. We investigate this exchange for OpenMP scheduling: how much tuning time and CPU consumption can be avoided by accepting a small loss in subsequent execution quality? A model trained on separate inputs ranks 12 LB4OMP schedulers. Budgeted screening measures only two, four, or eight candidates before committing; prediction-only, random screening, a trained fixed scheduler, and full 12-candidate search provide controls. We separately measure calibration cost, post-selection payload quality, and complete tuning-plus-useful-execution cost. The fresh experiment comprises 840 policy runs on 24 scenarios from three workload families, with five randomized blocks and directly measured process CPU seconds. Four-candidate screening saves 65.0% of tuning wall time and 65.9% of tuning CPU time, with a geometric mean payload penalty of 10.2% to the same-block empirical best. Its additional payload loss relative to full search's chosen schedule is -1.1% (95% CI -3.3%–0.9%). For 50 useful executions and an available prior, its total-cost ratio is 1.379 relative to full search (95% CI 1.329–1.435), where ratios above one favor screening. With calibration counted as useful work, the four-candidate wall ratios are 0.970 at 50 executions and 1.000 at 500, limiting the savings claim to the extra-calibration operating mode. The paper reports the complete budget–quality frontier, independent best-arm references, uncertainty, finite-horizon effects, and the number of deployments needed to amortize the prior. The result is a conditional answer to when reduced search is worthwhile, rather than a claim that a cheap selector must outperform the best scheduler itself.**

***Index Terms*—OpenMP, algorithm selection, tuning budget, CPU time, near-optimal scheduling, amortization, performance–cost trade-off.**

I. Introduction

An OpenMP scheduler assigns loop iterations to threads. Its execution quality depends on load imbalance, dispatch overhead, locality, and the current machine. Choosing the fastest scheduler can require executing the loop under several alternatives. That search is itself work. When an application is short, the time spent comparing schedules can exceed the time saved by the best schedule afterward.

The relevant engineering objective is therefore not always minimum post-selection loop time. An application may prefer a scheduler that is slightly slower per execution if it is substantially cheaper to identify. For example, avoiding 24 calibration executions can be useful even if the selected scheduler loses a few percent of throughput. Whether it is useful depends on the calibration cost, the execution horizon, the quality gap, and whether an offline prior must be paid for.

We study this decision directly. A trained model ranks the 12 schedulers in LB4OMP. Small candidate budgets restrict measurement to promising alternatives before committing to a fixed schedule. We compare the saved search time and process CPU seconds with the residual loss relative to an independently measured best fixed scheduler. Full search is the expensive comparator for tuning cost; the empirical best scheduler is the reference for execution quality. These roles are distinct.

This study follows an earlier local experiment that evaluated continuously updating bandits over 500 useful loop instances and did not establish an overall speed advantage. The present work asks a different question, uses fresh test inputs, separates calibration from useful execution, and directly measures CPU time. It is not a re-labeling of the earlier measurements.

Our research questions are:

1) How much calibration wall time and process CPU time does a restricted candidate budget save relative to a full 12-scheduler sweep?
2) What execution-speed penalty remains after the cheaper search, and how often does it stay within 5% or 10% of the empirical best reference?
3) For which useful-execution horizons does the tuning saving outweigh that penalty, and how many deployments amortize the offline prior?

The contributions are an explicit cost–quality formulation, an implemented budgeted selector in native C++, a new directly instrumented experimental corpus, and a deployment decision analysis including unfavorable cases. We claim neither a new bandit algorithm nor a guarantee of optimal scheduling.

II. Background and Positioning

OpenMP exposes standard schedule choices for shared-memory loops [1]. LB4OMP extends LLVM's OpenMP runtime with additional dynamic and adaptive schedulers [2]. Auto4OMP adds automated selection and an expert chunk-size calculation [3]. Our experiment reuses those implementations and the expert chunk rule; it changes the budget spent identifying a scheduler.

Müller Korndörfer et al. compare expert and reinforcement-learning selectors across six applications and three systems [4]. Their 12-scheduler explore-first formulation considers 144 ordered state–action combinations. They identify exploration

† *Code and data:* https://github.com/freebirdyeah/pcap_project_g3/

cost and reward choice as important practical issues. This motivates reducing selection effort, but does not require beating an offline best scheduler in subsequent loop speed. We instead expose the quality lost per unit of search work avoided.

Algorithm selection distinguishes an algorithm that is best on average from a virtual best choice for each instance. Tornede et al. note that exhaustive evaluation is costly and that feature-computation time must be included in runtime objectives [5]. This distinction applies directly here: a model that identifies a good schedule cheaply can be useful without identifying the fastest one every time.

Contextual bandits use context to predict action rewards [6]. Warm-start methods incorporate earlier information into their initial models [7]. We use a fitted disjoint linear model originating from that framework to rank candidates, then stop exploration at a declared budget. There is no online regret theorem or claimed confidence guarantee for this screening heuristic. The contribution is the measured budget–quality exchange.

The workload families follow the ideas of Mandelbrot, STREAM-style vector arithmetic, and sorted-adjacency triangle counting [8], [9]. Our kernels are original compact implementations, not the reference STREAM or GAP programs. Uncertainty is estimated at input-configuration level, following the need to distinguish repetitions from independent experiments in systems benchmarking [10].

III. Cost–Quality Formulation

Let $\mathcal{A}$ contain the 12 schedulers. A selector m spends tuning time S_m to commit to $a_m \in \mathcal{A}$. Afterward it executes H useful loop instances. With a reusable prior costing C_{prior} and shared across R deployments, the wall-time objective is

$$J_m^{\text{wall}}(H, R) = \frac{C_{\text{prior}}^{\text{wall}}}{R} + S_m^{\text{wall}} + \sum_{t=1}^{H} T_t^{\text{wall}}(a_m). \quad (1)$$

An analogous expression uses measured process CPU seconds. A selector without an offline prior, such as random screening or full search, has $C_{\text{prior}} = 0$ in this operational comparison. A fixed arm selected from the training campaign also uses offline information, so its acquisition cost must be charged when that information is not already available.

For an empirical best fixed-arm reference T^*, the residual payload slowdown is

$$\ell_m = \frac{\widetilde{T}(a_m)}{\widetilde{T}^*} - 1, \qquad \rho_m = \frac{\widetilde{T}^*}{\widetilde{T}(a_m)} = \frac{1}{1+\ell_m}, \quad (2)$$

where tildes denote medians of independent timing observations. ρ_m is retained speed relative to the reference. For example, a 5% runtime penalty retains approximately 95.24% of the reference speed. We study the frontier between S_m and ℓ_m, rather than treating reference-matching speed as the sole success criterion.

A. When is the exchange worthwhile?

Let F denote full search. For a cached prior, restricted selection is faster over horizon H whenever

$$S_F - S_m > \sum_{t=1}^{H} \left[T_t(a_m) - T_t(a_F)\right]. \quad (3)$$

Under stationary mean payload times and a positive gap $\Delta T = \bar{T}(a_m) - \bar{T}(a_F) > 0$, the estimated crossover is

$$H^* = \frac{S_F - S_m}{\Delta T}. \quad (4)$$

The restricted method benefits shorter executions and loses its advantage after sufficient repetition. If the measured gap is nonpositive, this model has no finite crossover favoring full search; that is an empirical case, not a promise about future executions. Finite search may choose a noisy winner, so full search is not assumed to be a perfect Oracle.

With an unpaid prior, the left side of Eq. 3 is reduced by C_{prior}/R. If $d(H)$ is the mean measured per-deployment saving, the minimum estimated reuse count is

$$R_{\min}(H) = \left\lceil \frac{C_{\text{prior}}}{d(H)} \right\rceil \quad \text{when } d(H) > 0. \quad (5)$$

If $d(H) \le 0$, repeated reuse does not amortize the prior for that comparison. Wall and CPU break-even counts are calculated separately. They are point estimates; an interval for mean savings that crosses zero does not support a confident finite amortization claim.

B. Two meanings of exploration

The primary experiment uses *tune-then-run*: calibration executions are additional evaluations before the H useful loop instances. This is suitable for repeated kernels with representative trial inputs or a calibration phase whose results are not counted as application progress. It is the setting in which fewer evaluations directly avoid extra work.

In an in-situ selector, exploratory loop instances may already advance the application. Removing 24 trials does not then remove 24 required useful executions. Its cost is the excess time of those trials relative to a deployment schedule, not their full execution time. The present measurements do not establish equivalent savings in that setting. This semantic distinction is necessary to interpret the proposed trade-off.

IV. Budgeted Candidate Screening

A. Prior and context

The context is $x = [1, \log_2 N, \log_2 P, \log_2 T_0, L_0, V_0]^\top$, where T_0 is the median of three STATIC probes, L_0 is median finish-time imbalance, and V_0 is median coefficient of variation of thread finish times. Thread finishes are measured relative to parallel-region start, so launch and arrival effects are included. Non-intercept features are standardized using training data.

For arm a, a ridge model is initialized using training scenario contexts and normalized log-time rewards:

$$r_{i,a} = \mathrm{clip}\left(-\frac{\log(T_{i,a}/T_{i,\mathrm{STATIC}})}{\log 4}, -1, 1\right), \tag{6}$$

$$A_a = \lambda I + w \sum_i x_i x_i^\top, \tag{7}$$

$$b_a = w \sum_i x_i r_{i,a}, \qquad \hat{\theta}_a = A_a^{-1} b_a. \tag{8}$$

Arms are ranked by $x^\top \hat{\theta}_a$. The inherited values $w = 0.25$ and $\lambda = 1$ are frozen from the preceding study's training-only validation. A stored exploration coefficient of 0.025 is not used in the greedy ranking. This paper tests fixed measurement budgets, not continuously updating LinUCB.

B. Budget definitions and controls

Table I defines the seven treatments. The three STATIC probes also supply a measured safety candidate. For Guided q, we test the top $q-1$ predicted non-STATIC arms three times each, then choose the minimum observed median among those arms and STATIC. The cap is an explicit number of candidates, not a claim that each candidate has equal execution cost.

Full search saves no calibration evaluations relative to itself. Guided 4 reduces the count by two thirds; Prior only reduces it by 91.7%. These exact count reductions do not imply equal wall-time or CPU reductions. Actual selection costs are measured, and residual payload quality is evaluated independently after commitment. No method continues switching during payload.

V. Implementation and Experimental Protocol

A. Native implementation and CPU accounting

The C++ implementation reuses the previously audited workload and scheduling code. It adds explicit probe, tuning, and payload phases. Wall time uses a monotonic clock. CPU time uses `CLOCK_PROCESS_CPUTIME_ID`, which accounts for CPU time consumed by all threads in the process [11]. It includes active waiting within the OpenMP runtime during the measured phase. It is neither energy consumption nor a count of FLOPs or instructions.

Each timed loop records wall seconds, process CPU seconds, scheduler ID, kernel elapsed time, imbalance, and variation. Prefix snapshots record actual wall and CPU costs for the first 10, 50, 100, and 500 useful executions. These are measured prefixes of a committed run, not independently repeated experiments at four horizons. Model deserialization and ranking are charged to tuning for prior-based methods. Common input construction and two STATIC warm-up executions are excluded from deployment cost for all treatments. Whole-process elapsed and CPU costs are also recorded.

The machine is an AMD Ryzen 5 7235HS with four physical cores/eight logical CPUs and approximately 7.6 GiB visible RAM under WSL2 Linux `6.6.87.2`. Clang 16.0.6 uses `-O3 -std=c++17 -fopenmp -ffp-contract=off`. LB4OMP commit `b9d25eca7265a4b5b3a675a1765d489775847938` has a portability-only initialized-array-to-vector patch. Threads are pinned to physical-core representatives $0, 2, 4, 6$; the active wait policy and 200 ms block time are fixed. Dynamic teams are disabled. Host activity, thermal state, and turbo are uncontrolled and limit the precision of small differences.

The expert chunk rule is enabled for every treatment. In the pinned runtime, it can override STATIC's chunk size as well; STATIC therefore uses cyclic expert-sized chunks rather than a single contiguous block per thread. The scheduling portfolio is STATIC, SS, TSS, AutoLLVM, GSS, mFAC2, Static Steal, AWF-B, AWF-C, AWF-D, AWF-E, and mAF.

B. Fresh training and unseen test inputs

We collect fresh calibration measurements for the original training configurations, seeds 0 and 1, at two and four threads. For every scenario, all 12 arms run in three separate processes. Each process has three STATIC probe observations and 20 observations of the target arm. One median label per scenario/arm fits the model, rather than treating individual loop rows as independent prior evidence. This is 432 calibration processes. The training-selected fixed arm is SS.

New test configurations use seeds 4–7. They are disjoint from both training and the preceding paper's seeds 2 and 3. Table II lists all sizes. Each configuration runs at $P \in \{2, 4\}$, yielding 24 scenarios and 12 input configuration clusters. All arms and treatments within one scenario use identical deterministic inputs and produce the same useful outputs.

Mandelbrot uses a square row-major grid, a real-coordinate shift of $0.025(\mathrm{seed} \bmod 4)$, and a seed-dependent iteration cap. Triad evaluates $a_i = b_i + 3c_i$ on three double arrays. The graph generator makes $24N$ undirected edge proposals using a seeded `mt19937`; one third have an endpoint in a hub-biased prefix, while others are uniform. It removes self-edges and duplicates, orients by vertex ID, and counts triangles using sorted-neighbor intersections. Graph construction is outside tuning and payload timing. These small synthetic inputs do not establish performance on large real graphs or production HPC applications.

C. Randomized blocks and independent references

The frozen protocol declares five blocks. Every block shuffles 168 policy processes and 288 independent fixed-arm reference processes. Each reference process measures 25 payload executions of one arm after the common warm-ups; it does not provide labels to the selector. Reference and policy jobs execute sequentially to avoid concurrent benchmark contention. Source, binary, model, protocol, raw CSV, and summary hashes are recorded.

For scenario s and block b, the primary empirical quality reference is

$$\widetilde{T}^*_{s,b} = \min_{a \in \mathcal{A}} \mathrm{median}_{j=1}^{25}\, T^{\mathrm{reference}}_{s,b,a,j}. \tag{9}$$

The selected policy's quality uses its median over 500 post-selection payload observations in a separate process. These loop observations are not treated as independent experimental

TABLE I
POLICIES AND EXTRA CALIBRATION EXECUTIONS. MODEL RANKING AND LOADING COST ARE INCLUDED IN TUNING TIME WHERE USED.

Policy	Candidates tested	Evaluations	Commitment rule
Best fixed	0	0	Training-best arm; no online tuning.
Prior only	1	3	Three STATIC probes; choose the prior's best predicted arm, which may not itself have been measured online.
Guided 2	2	6	STATIC plus the top predicted non-STATIC arm.
Guided 4	4	12	STATIC plus three predicted candidates.
Guided 8	8	24	STATIC plus seven predicted candidates.
Random 4	4	12	STATIC plus three uniformly sampled non-STATIC arms.
Full 12	12	36	All arms, three observations each; minimum median.

TABLE II
NEW HELD-OUT CONFIGURATIONS, EACH MEASURED AT TWO THREAD COUNTS.

	Seed 4	Seed 5	Seed 6	Seed 7
Mandelbrot pixels	36,864	57,600	20,736	43,264
Maximum pixel iterations	192	216	240	264
Triad elements	524,288	655,360	458,752	917,504
Triangle vertices	768	1,792	1,280	1,920

replications. A minimum over noisy reference medians is optimistic. We do not treat this value as a known mathematical optimum, and negative observed slowdowns are retained as noise rather than clipped.

A cross-fitted sensitivity selects the reference arm using the other four blocks, then evaluates that arm's time in the held-out block. This separates selecting the reference arm from measuring its held-out time. It is an analysis-only addition recorded during the first block, before aggregate test outcomes were examined. Both reference definitions are reported.

D. Correctness and uncertainty

The shared kernels passed a scheduler-coverage test for all 12 arms at both thread counts and an independent cubic triangle-count check on a tiny graph. Every new process compares its final parallel checksum with a serial evaluation. Candidate counts, selected-arm identity, phase lengths, and raw hashes are audited. Repeated identical input means final-output checks alone do not certify each intermediate iteration; coverage is tested separately.

Paired effects compare policies within the same scenario and block. We report geometric mean ratios and 95% bootstrap intervals from 10,000 resamples of the 12 input configurations, with paired observations resampled inside each cluster. Both thread counts belong to their common input cluster. Individual loop observations and horizon prefixes are not independent replications. Acceptance rates within 5% and 10% are descriptive empirical frequencies, not validated deployment guarantees.

VI. RESULTS: THE SEARCH–QUALITY FRONTIER

The primary corpus contains 2,712 audited processes and 477,096 logged loop observations, including calibration and references. There are 840 policy runs and 420,000 useful online executions. All process output checks passed. The following results separate selection saving, payload quality, and complete deployment cost.

A. Calibration time and CPU savings

Guided 2, Guided 4, and Guided 8 save 82.4%, 65.0%, and 31.9% of tuning wall time, respectively. Their corresponding CPU savings are 83.2%, 65.9%, and 32.6%. Prior-only selection saves 91.1% wall time and 92.5% CPU time. These are paired geometric cost ratios over complete tuning phases, not ratios of median milliseconds. The evaluation-count reductions are fixed by policy; measured wall and CPU reductions need not equal them because candidates have different costs and model loading adds overhead. CPU seconds include runtime active waiting, so they quantify process resource consumption under the stated runtime configuration, not an architecture-independent operation count.

B. Residual scheduling quality

Guided 4's geometric mean payload loss is 10.2% (95% CI 7.5%–13.3%). It retains 90.7% of reference speed. Relative to the schedule actually selected by Full 12, its extra payload loss is -1.1% (95% CI -3.3%–0.9%). Its payload is within 5% in 44.2% of runs and within 10% in 64.2%. Full search itself has a geometric residual of 11.4%, illustrating the gap between a finite noisy sweep and an independent best-arm reference. Figure 1 places measured tuning saving and residual payload loss on separate axes. A useful policy moves toward lower search cost while keeping its payload loss within an application's declared tolerance.

C. Does the saved tuning work repay the quality loss?

Guided 4's wall-cost ratios at horizons 10, 50, 100, and 500 are 2.014, 1.379, 1.200, 1.053. At 50 executions the CPU-cost ratio is 1.384 (95% CI 1.335–1.440); at 500 it is 1.053. Ratios above one indicate lower total cost for screening. These ratios

TABLE III

MEASURED TUNING COST RELATIVE TO FULL 12. SAVINGS ARE PAIRED GEOMETRIC EFFECTS; MEDIAN WALL MILLISECONDS ARE DESCRIPTIVE.

Policy	Evaluations	Median wall (ms)	Wall saved (%)	CPU saved (%)
Best fixed	0	0.01	100.0	100.0
Prior only	3	8.24	91.1	92.5
Guided 2	6	16.45	82.4	83.2
Guided 4	12	32.16	65.0	65.9
Guided 8	24	63.48	31.9	32.6
Random 4	12	30.91	67.2	67.7
Full 12	36	91.14	0.0	0.0

TABLE IV

INDEPENDENT PAYLOAD-QUALITY MEASUREMENTS. GEOMETRIC MEAN AND 90TH-PERCENTILE RESIDUAL SLOWDOWNS ARE RELATIVE TO THE SAME-BLOCK EMPIRICAL BEST. ACCEPTANCE RATES ARE OBSERVED RUN FRACTIONS.

Policy	Mean loss (%)	P90 loss (%)	Within 5%	Within 10%
Best fixed	10.2	32.9	40.0	65.8
Prior only	10.4	32.4	41.7	66.7
Guided 2	9.9	33.1	36.7	72.5
Guided 4	10.2	31.2	44.2	64.2
Guided 8	10.4	26.8	38.3	67.5
Random 4	11.2	35.2	35.0	62.5
Full 12	11.4	36.0	40.8	65.8

TABLE V

TOTAL WALL-COST RATIOS: FULL 12 COST DIVIDED BY POLICY COST, INCLUDING TUNING AND H USEFUL EXECUTIONS. VALUES ABOVE ONE FAVOR THE POLICY. PRIOR ACQUISITION IS EXCLUDED HERE AND CHARGED SEPARATELY.

Policy	H=10	H=50	H=100	H=500
Best fixed	4.556	1.704	1.360	1.084
Prior only	3.453	1.621	1.333	1.081
Guided 2	2.795	1.544	1.294	1.069
Guided 4	2.014	1.379	1.200	1.053
Guided 8	1.327	1.166	1.104	1.032
Random 4	2.128	1.403	1.226	1.050
Full 12	1.000	1.000	1.000	1.000

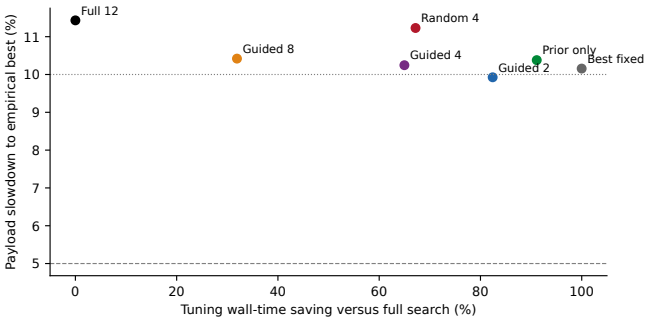


Fig. 1. Measured tuning wall-time saving versus residual payload slowdown. Points are geometric aggregate effects over all scenarios and blocks. Horizontal guides mark 5% and 10% slowdown tolerances; they are application preferences, not statistical confidence limits.

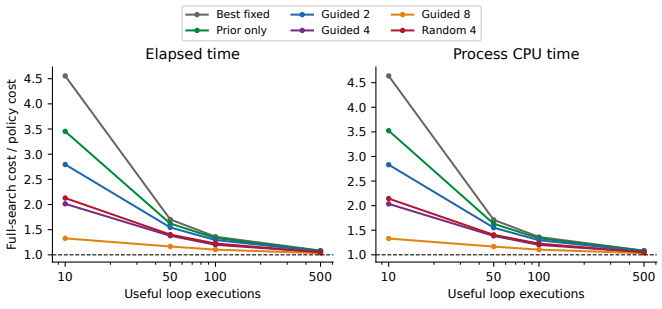


Fig. 2. Full-search deployment cost divided by each policy's deployment cost, including tuning. Values above one favor the restricted policy. The left panel measures elapsed time; the right measures process CPU time. Horizon prefixes come from the same committed runs.

include calibration and the same number of useful executions. They assume the prior is already available; its acquisition is charged in the following subsection. A method can save most tuning work yet lose its total-cost advantage over long horizons when its committed arm is slower.

At 50 useful executions, 44.2% of Guided 4 runs satisfy (95% cluster-bootstrap interval 27.5%–60.8%) all three conditions of lower wall cost, lower CPU cost, and at most 5%

payload loss; with a 10% tolerance this rises to 64.2%. At 500 executions the corresponding rates are 42.5% and 60.0%. The joint criterion requires lower total wall time, lower total CPU time, and payload loss within the stated tolerance in the same observed run. It is more demanding than reporting either tuning saving or quality alone. For Guided 4, 64 of 120 paired runs have both a tuning saving and a positive payload mean gap to full search. Their median estimated finite crossover is 935 useful executions. This summarizes only runs with a positive gap; it is not a universal operating cutoff.

D. The cost of acquiring the prior

Fresh operational calibration and fitting cost 35.97 wall seconds and 83.89 process CPU seconds. Calibration includes complete subprocess costs—input construction, warm-ups, timing, correctness validation, and output—while fitting includes loading measurements, computing statistics, and serializing the model. Build time, Python import startup, and the earlier research campaign that chose inherited hyperparameters are excluded. The reported cost therefore describes operational reuse of the fixed recipe, not the entire cost of inventing the method.

For Guided 4 at horizon 50, the estimated break-even is 488 deployments for wall time and 409 for CPU time. At horizon 500 the estimates are 531 and 448, respectively. When both resources matter, the larger finite count is the relevant lower bound. Equation 5 uses arithmetic mean per-deployment savings because actual seconds, rather than logarithmic ratios, repay an upfront expense. The reuse counts are estimates for the measured workload mixture. If the mean saving is nonpositive, no finite count is reported. A prior available from an existing deployment and a newly acquired prior are materially different operating conditions.

E. Reference sensitivity and simple alternatives

Under the cross-fitted reference, Guided 4's geometric payload loss becomes -1.7%; its within-5% and within-10% rates become 85.0% and 92.5%. The difference from the primary reference quantifies sensitivity to choosing a minimum among noisy same-block timings. The cross-fitted reference reduces same-block winner selection bias but is not a guarantee of the true optimum either. Timing noise, family similarity, and the limited number of configurations still affect both estimates.

Random 4 has a geometric payload loss of 11.2% and a horizon-50 wall-cost ratio of 1.403. The trained fixed arm has a loss of 10.2% and ratio 1.704, with no online calibration. Thus the value of prediction must be considered alongside the simpler ways to spend less on search. A zero-online-tuning fixed arm can be a strong choice when the training distribution is representative. Random screening tests whether candidate count alone explains a favorable trade-off. These controls prevent attributing every saved evaluation to contextual prediction.

F. Secondary equal-useful-work experiment

To examine the original online interpretation directly, we additionally measure Prior only, Guided 4, and Full 12 with exactly 50 or 500 total useful executions, counting calibration calls as useful. For $H = 50$, Full 12 spends 36 useful calls measuring candidates and then performs 14 committed calls; Guided 4 spends 12 and then performs 38. Thus both do the same total work. These are new direct measurements in three randomized blocks on all 24 scenarios, not reconstructed prefixes of the primary experiment.

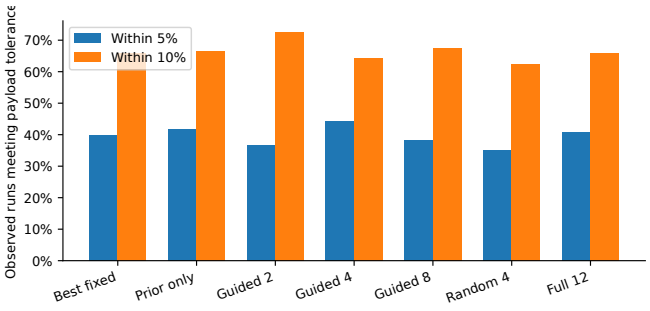


Fig. 3. Observed policy runs whose payload median is within 5% or 10% of the independent same-block empirical best. These rates measure execution quality separately from search saving and do not imply a probabilistic guarantee for new programs.

The executable and fitted model are unchanged. Each native process is given $H - n_m$ committed payload calls, where n_m is its calibration count, and we charge the measured tuning and complete payload phases. The additional experiment was declared during the primary campaign, before aggregate outcomes were inspected. Its 432 processes contain 118,800 useful executions. For 50 equal useful executions, Guided 4's wall-cost ratio is 0.970 (95% CI 0.933–0.999) and its CPU-cost ratio is 0.970 (95% CI 0.932–1.001). For 500 equal useful executions, Guided 4's wall-cost ratio is 1.000 (95% CI 0.987–1.013) and its CPU-cost ratio is 1.001 (95% CI 0.987–1.014).

VII. Discussion: Is the Trade-off Worth It?

For the four-candidate policy, the measured exchange is 65.0% less tuning wall time and 65.9% less tuning CPU time for 10.2% aggregate residual payload loss. Its horizon-50 total wall ratio is 1.379, whereas its horizon-500 ratio is 1.053. These measurements make the trade-off explicit; the tolerance hit rate and prior cost determine whether it is acceptable for a particular deployment.

The equal-useful-work experiment qualifies this conclusion for in-situ use: the four-candidate wall-cost ratios are 0.970 at 50 useful calls and 1.000 at 500. These do not establish a time/CPU saving from reduced search in that mode. Avoided candidate tests cannot be counted as avoided necessary executions; the primary extra-calibration savings apply only when calibration is additional work.

The objective is a choice among resource and quality preferences. A user who accepts a small loss in per-loop speed may prefer cheap screening when the run is short or the prior is already available. A long, repeated execution can justify more measurement because its payload dominates the total cost. An

TABLE VI
FOUR-CANDIDATE SCREENING BY WORKLOAD FAMILY: CALIBRATION WALL SAVING, PAYLOAD LOSS TO THE EMPIRICAL BEST, AND COMPLETE DEPLOYMENT WALL-COST RATIOS. EACH FAMILY CONTAINS FOUR INPUT CONFIGURATIONS AT TWO THREAD COUNTS.

Family	Tuning saved (%)	Payload loss (%)	H=50 ratio	H=500 ratio
Mandelbrot	64.6	7.5	1.369	1.034
Triad	63.0	11.4	1.351	1.076
Triangle	67.2	12.0	1.418	1.050

TABLE VII
SECONDARY EQUAL-USEFUL-WORK RESULTS. BOTH TUNING AND COMMITTED CALLS COUNT TOWARD H; COST RATIOS ARE FULL 12 DIVIDED BY THE POLICY. ALL INTERVALS USE INPUT-CLUSTER RESAMPLING.

Policy	H	Wall ratio	Wall 95% CI	CPU ratio
Prior only	50	0.990	0.950–1.026	0.991
Guided 4	50	0.970	0.933–0.999	0.970
Prior only	500	0.997	0.978–1.015	0.998
Guided 4	500	1.000	0.987–1.013	1.001

expensive prior acquired for a single short deployment can erase the online saving. The correct comparison is Eq. 1, with a declared horizon, tolerance, and reuse assumption.

Our implementation makes budget decisions explicit: measure only a declared candidate set, stop, and commit. The quality penalty is therefore attributable to that selection rule and measurement noise, rather than being hidden inside continuous online exploration. However, candidate count is only a proxy for resource use. The directly measured wall and CPU costs are the evidence for saving time and compute in this experiment.

Those savings do not establish that contextual prediction is necessary. The fixed and random-screening controls also reduce search effort; their results must remain visible when assessing the learned ranking. Nor does spending a larger budget ensure a strictly better committed schedule in every finite noisy run. The measured exchange is about operational cost and execution quality, not an ordering theorem over candidate counts.

The full sweep may itself choose a noisy winner from only three observations per arm. Its payload need not match the independent best-arm reference. This is why we report both a costly feasible search comparator and an independent execution-quality reference. Near-reference quality with lower search cost is the target; beating an unavailable zero-cost Oracle is not.

VIII. LIMITATIONS AND SCOPE

Tuning semantics. Savings are measured for additional calibration executions. They do not automatically transfer to in-situ application progress. The method assumes repeatable representative probes and an input state that can be reused or restored cheaply. Stateful applications may have additional copying or reset costs that are absent from these kernels.

Machine and workloads. One four-core WSL2 system and compact synthetic kernels do not support conclusions about multi-socket HPC, NUMA, heterogeneous devices, or large real-world graphs. Active waiting affects CPU consumption. Host activity, frequency, and temperature are uncontrolled. Training and test inputs belong to the same three families; seed holdout is not unseen-application generalization.

Reference and tolerance. Finite reference medians and their minimum are noisy. Threshold rates near 5% can change under reference uncertainty. The cross-fitted sensitivity addresses one source of optimism but not all nonstationarity. No worst-case near-optimality guarantee is claimed. References contain 25 observations per process, whereas committed policy quality uses 500. Different run duration can expose cache, adaptive-runtime, frequency, or thermal transients. We cannot isolate those effects here. Negative cross-fitted gaps do not show that a policy beats a true optimum; they show that a finite empirical reference can be outperformed or fluctuate.

Prior and policy design. Hyperparameters are inherited rather than retuned for this budgeted objective. Three repeats per candidate are fixed. Rank-based screening can omit a useful arm, and the model's clipped log-time target is a surrogate rather than an exact total-cost objective. Tolerance and budget are fixed preferences, not adaptively chosen per deployment.

Statistical and cost scope. There are 12 input clusters, not hundreds of thousands of independent experiments. Prefixes share observations. Crossover and amortization extrapolate stationary means. Operational training cost excludes method-development effort. CPU time measures consumed processor time, including waiting, and cannot establish energy or instruction savings.

IX. CONCLUSION

We evaluate OpenMP scheduler selection as a budget–quality decision. The implementation directly measures the work spent choosing a schedule and the execution quality left after that choice. On the measured four-core system, Guided 4 avoids two thirds of calibration evaluations and saves 65.0% tuning wall time and 65.9% tuning CPU time. Its geometric payload penalty is 10.2%. For an available prior, its total wall-cost ratio is 1.379 at 50 useful executions and 1.053 at 500. Prior acquisition and tail losses remain part of the

decision rather than being hidden costs. The separate equal-useful-work ratios of 0.970 and 1.000 do not establish an in-situ benefit. The practical decision should declare a useful-execution horizon, a tolerated payload loss, whether calibration is extra work, and how the prior is paid for. Reporting those conditions makes a near-best scheduling policy useful on its own terms without requiring it to be the fastest scheduler.

REFERENCES

[1] OpenMP Architecture Review Board. OpenMP Application Programming Interface, Version 5.2 (2021). https://www.openmp.org/specifications/

[2] J.H. Müller Korndörfer, A. Eleliemy, A. Mohammed, F.M. Ciorba. LB4OMP: A Dynamic Load Balancing Library for Multithreaded Applications. IEEE TPDS 33(4), 830–841 (2022). https://doi.org/10.1109/TPDS.2021.3107775

[3] A. Mohammed, J.H. Müller Korndörfer, A. Eleliemy, F.M. Ciorba. Automated Scheduling Algorithm Selection and Chunk Parameter Calculation in OpenMP. IEEE TPDS 33(12), 4383–4394 (2022). https://doi.org/10.1109/TPDS.2022.3189270

[4] J.H. Müller Korndörfer, A. Mohammed, A. Eleliemy, Q. Guilloteau, R. Krummenacher, F.M. Ciorba. A Comparative Study of OpenMP Scheduling Algorithm Selection Strategies. IEEE Access 13, 151216–151234 (2025). https://doi.org/10.1109/ACCESS.2025.3602234

[5] A. Tornede, L. Gehring, T. Tornede, M. Wever, E. Hüllermeier. Algorithm Selection on a Meta Level. Machine Learning 112, 1253–1286 (2023). https://doi.org/10.1007/s10994-022-06161-4

[6] L. Li, W. Chu, J. Langford, R.E. Schapire. A Contextual-Bandit Approach to Personalized News Article Recommendation. In WWW, 661–670 (2010). https://doi.org/10.1145/1772690.1772758

[7] B. Oetomo, R.M. Perera, R. Borovica-Gajic, B.I.P. Rubinstein. Cutting to the Chase with Warm-Start Contextual Bandits. Knowledge and Information Systems 65, 3533–3565 (2023). https://doi.org/10.1007/s10115-023-01861-2

[8] J.D. McCalpin. STREAM: Sustainable Memory Bandwidth in High Performance Computers. https://www.cs.virginia.edu/stream/

[9] S. Beamer, K. Asanović, D. Patterson. The GAP Benchmark Suite. arXiv:1508.03619 (2015). https://arxiv.org/abs/1508.03619

[10] T. Kalibera, R. Jones. Rigorous Benchmarking in Reasonable Time. ISMM, 63–74 (2013). https://kar.kent.ac.uk/33611/

[11] Linux man-pages project. `clock_gettime(3)`: clock and process CPU-time definitions. https://man7.org/linux/man-pages/man3/clock_gettime.3.html